

Andrea Brilli

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📍 Sapienza, University of Rome, Department of Computer, Control and Management Engineering "Antonio Ruberti"

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Education

Sapienza, University of Rome , Operations Research	Rome, Italy
- Curriculum: Operations Research - Thesis: Derivative-Free Optimization: worst-case complexity for Line-Search methods and a mixed Penalty-Barrier approach - ABRO (Automatic control, Bioengineering and Operations Research)	2021 – 2025
Sapienza, University of Rome , Management Engineering	Rome, Italy
- Curriculum: Decision Models - Thesis: An interior penalty method for black-box constrained optimization	2019 – 2021
Sapienza, University of Rome , Management Engineering	Rome, Italy
	2015 – 2019

Experience

Sapienza University of Rome , PostDoc	Rome, Italy
Department of Computer, Control and Management Engineering. Funded by PRIN (NextGenerationEU funds) for hydrodynamic optimization study on biomimetic and innovative hydrodynamic configurations for autonomous underwater gliders (AUG), specifically designed for ocean surveying.	2024 – present
- Support to PhD students on advanced techniques for training neural networks and derivative-free techniques - Collaboration with INESC MN (Lisbon, Portugal) for optimization of structural design parameters of metamaterials - Collaboration with NOVA University of Lisbon on techniques to approximate the Pareto Front of constrained multiobjective optimization problems - Collaboration with Polytechnique Montréal to study nonsmooth constrained optimization problems - Collaboration with University of Florence to enhance performance of Augmented Lagrangian techniques	2 years
European Space Agency - ESRIN - ϕ-lab , Research Visiting	ESRIN, Italy
Collaboration to develop denoising techniques for satellite data striping noise.	2026 – present
	1 year
Polytechnique Montréal , Ph.D. Visiting Researcher	Montréal, Canada
Six months of collaboration with Professor Sébastien Le Digabel and Professor Youssef Diouane to enhance the performance of NOMAD solver.	2024 – 2024
	1 year
Universidade NOVA de Lisboa , Ph.D. Visiting Researcher	Lisbon, Portugal
One year of collaboration with Professor Ana Luísa Custódio and Ph.D. Everton Jose da Silva to enhance the performance of the SID-PSM algorithm.	2022 – 2023
	1 year

Teaching

Sapienza University of Rome , Bachelor degree in Computer Science Engineering	Rome, Italy
Introduction to multivariate calculus and mathematical optimization	Sept 2025 – Jan 2026
Sapienza University of Rome , Master degree in Industrial Engineering	Rome, Italy
Advanced algorithms for nonlinear optimization	Mar 2026 – June 2026

Awards

AIROYoung Dissertation Award

Sept 2025

Yearly award for the best doctoral thesis in Operations Research defended in Italy.
AIROYoung (Italian Operations Research Society)

Publications

Complexity results and active-set identification of a derivative-free method for bound-constrained problems

Accepted for publication in JOTA. Analysis of complexity and active-set identification for derivative-free optimization with bound constraints.

A. Brilli, A. Cristofari, G. Liuzzi, S. Lucidi

arxiv.org/abs/2402.10801

A penalty-interior point method combined with mads for equality and inequality constrained optimization

Novel constraint-handling method combining penalty and interior point approaches with MADS algorithm.

C. Audet, A. Brilli, Y. Diouane, S. Le Digabel, E. Silva, C. Tribes

arxiv.org/abs/2601.20811

Tunable ito-metal plasmonic metamaterial channel for tailored sensing: A simulation-driven approach

Optimization-driven design of plasmonic metamaterial sensors for tailored sensing applications.

C. Alfisi, A. Brilli, H. Terças, S. Cardoso

doi.org/10.1063/5.0297834

An interior point method for nonlinear constrained derivative-free optimization

Novel interior point approach for solving general nonlinear constrained problems without derivatives.

A. Brilli, G. Liuzzi, S. Lucidi

doi.org/10.1080/10556788.2025.2453110

Nonlinear derivative-free constrained optimization with a mixed penalty-logarithmic barrier approach and direct search

Mixed penalty-barrier method combined with direct search for general constrained derivative-free optimization.

A. Brilli, A. L. Custódio, G. Liuzzi, E. J. Silva

arxiv.org/abs/2407.21634

Worst case complexity bounds for linesearch-type derivative-free algorithms

Theoretical complexity analysis for linesearch-based derivative-free optimization methods.

A. Brilli, M. Kimiaiei, G. Liuzzi, S. Lucidi

doi.org/10.1007/s10957-024-02519-x

Skills

Optimization

Programming

Software Development

Languages

Italian

Native speaker

English

Fluent

Portuguese

Basic

Spanish

Basic

Certificates

Deep Learning Specialization	2020
Data-driven Astronomy	2020
CyberChallenge 2020 Training programme organized by CINI in collaboration with the Ministry of Defence and SISR.	2020

Projects

DFL - Derivative-Free Library Collaborator for the implementation of algorithms in Python within the DFL library. Author of LOG-DFL.	2021 – present
LOG-DS Co-author of the LOG-DS algorithm, a direct search method for general nonlinear constrained problems implemented in Matlab.	2023 – present
NOMAD A Blackbox Optimization Software. Supported the implementation of new constraint-handling strategies within the package.	2024 – present

References

Professor Giampaolo Liuzzi**Professor Stefano Lucidi****Professor Ana Luísa Custódio****Professor Sébastien Le Digabel****Professor Youssef Diouane**